Bihar Engineering University, Patna
End Semester Examination – 2023
Semester-IV

Course: B.Tech. *Time: 03 Hours*

Code: 104404 *Subject: Electromagnetic Theory* *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 ***Answer any seven question only:*** **[2 x 7 = 14]**

(a) What is physical significance of divergence?
(b) State Stoke's theorem.
(c) How is the unit vectors defined in three co ordinate systems?
(d) State Gauss law for electric fields.
(e) Define dielectric strength.
(f) State Biot –Savarts law.
(g) What is magnetic susceptibility?
(h) Write down the wave equation for E and H in free space.
(i) State Poynting Theorem.
(j) Define intrinsic impedance or characteristic impedance.

Q.2 (a) Derive the expression for the attenuation constant, phase constant and intrinsic impedance for a uniform plane wave in a good conductor. **[7]**

(b) Calculate the depth of penetration in copper at 10 MHZ. Given the conductivity of copper is 5.8×10^7 S/m and its permeability 1.3 mH/m. **[7]**

Q.3 (a) Derive suitable relations for integral and point forms of poynting theorem. **[7]**

(b) Discuss about the plane waves in lossless dielectrics. **[7]**

Q.4 With necessary explanation, derive the Maxwell's equation in differential and integral forms. **[14]**

Q.5 (a) An iron ring with a cross sectional area of 3 cm^2 and mean circumference of 15 cm is wound with 250 turns wire carrying a current of 0.3A.The relative permeability of ring is 1500. Calculate the flux established in the ring. **[7]**

(b) Derive the expressions for boundary conditions in magnetic fields. **[7]**

Q.6 (a) State and proof gauss law and explain applications of Gauss law. **[7]**

(b) Explain Poisson's and Lapace's equations. **[7]**

Q.7 (a) State and proof divergence theorem. **[7]**

(b) Define divergence, gradient, curl in spherical co-ordinate system with mathematical expression **[7]**

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Q.8 (a) Define and explain Biot –Savart Law . [7]

(b) Derive General field relation for time varying electric and magnetic fields using Maxwell's' equations. [7]

Q.9 Write short notes on *any two* of the following:- [7 x 2 =14]

(a) Equipotential Surface.

(b) Magnetic Vector potential.

(c) UHF Transmission line.

(d) Smith Chart.

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